Brief 9 — siena results, Wave 0 and stage 1 sweep complete

11 November 2026, 05:00 GMT · commit 6cb459c (task 6: cc83a62 = 6cb459c + preliminary protocol) (local branch audit-brief9 = 8fa4b74 + Brief 8 and 9 patches, clean tree, not pushed) · RTX 5090 machine · pool md5 341e229d88f4 · validation pseudo-folds with clean context only; nothing scored on the UCLH test folds · aggregates only.

> **Gates:** patched scorer reproduces final1M seed 0 exactly (0.000 both splits); old leave-out re-estimate +0.0028 [−0.0012, +0.0067] as reported; 23 new tests pass (full suite 416 passed, 4 failed; the same 4 fail on unpatched 8fa4b74).

> **Not started:** tasks 7 (new code) and 8 (rival fine-tuning). Task 6 ran under a preliminary protocol that César has not yet reviewed.

Decisions for César

- **Task 1** (Item 0a: allocation types, near-copies, latent R^2, distinct contexts): (1) the "allocation off" axis does not run: 0.06 % agnostic processes survive the verifier (rule: > 1 %); (2) the verifier rejects agnostic draws at zero confounding strength, so 3.5.2 should say the trained prior has practically no agnostic (U-driven) allocation; (3) DISTINCT=1 cannot make contexts distinct above the slice size (~780 rows): over 1,024 rows 99–100 % of contexts are longer than their slice.
- **Task 9** (Proximity check (stage 2 preparation)): the threshold (fixed in advance by the plan). Two benchmark trials have a virtual network of the prior at $|r| \geq 0.9$; sixteen at ≥ 0.8.
- **Task 5a** (KCH split checks (Brief 9 point 8)): whether a 604-lesion test half is acceptable. No B-half trial has fewer than 50 enrolled (minimum 50, median 142), but 14 of 32 B-half trials have a class (only-half-1 or only-half-2) below 10, against 8 of 32 in the whole pool; 6 of those classes are structurally 0 in the whole pool too.
- **Task 5b** (R0 trained on KCH half A, scored on validation (Brief 9 point 8, task 5)): whether to adopt the KCH split as the fresh test. Training R0 on half the pool (603 lesions) costs 0.0025 PEHE against final1M (interval [−0.0051, +0.0001], p 0.058). That is below the 0.005 screening rule and not significant, but the sign is consistent across splits. Together with 5a (thin hit-classes in 14 of 32 B-half trials), the split looks usable for PEHE but weak for prescriptive balanced accuracy.
- **Task 4** (Learning curve of the latent reference (validation only)): none yet. On the only cells large enough to compare (20 matched cells: network 10 in both splits × 10 pseudo-folds), the reference is still falling between 2,000 patients and all, but by less each doubling; the full table is in learning_curve_summary_E1.csv.
- **Task 2** (Wave 0 item 1: nine cells on validation, with pehe_implied): (1) in the noisy cells, against the implied CATE $TE(1 - RE)\cdot\tau$, the final model is clearly worse than TabPFN-3.5 and CausalPFN (+0.10 to +0.17 PEHE): it does not shrink its effects as much as the noise requires, so the direction left open in Brief 8 (explanation 5) is "the final model overshoots"; (2) under location bias (0.95 and the default) the final model beats every estimator by 0.05–0.07, the result that supports the biased-cell criterion; (3) the primary biased cell to fix in the protocol (location bias 0.5 proposed) is the weakest one: at 0.5 TabPFN is still 0.014 better.
- **Task 6** (Stage 1 sweep: 2^(4−1) factorial and generic share 0.25 (24 runs), screened against R0): (1) no level passes the screen, so under the preliminary protocol the next prior stays at R0 (final1M); (2) wide virtual-trial ranges (factor B) cost 0.013 PEHE, a robust harm in every run that has them; (3) generic share 0.75, effect term 0 and cosine decay are flat ($|effect| \leq 0.0014$, none significant); (4) the preliminary protocol (docs/protocol_stage1.md, "preliminary") needs your review, the rotation-averaging point first.
- **Task 3** (Leave-out at 1M with three seeds: held vs seen, run as a random effect (Brief 9 point 2)): none needed to proceed. Leaving a network out of the prior costs nothing measurable on it: held − seen = +0.0004 PEHE, REML interval [−0.0015, +0.0023]. A penalty above ~0.002 for unseen

networks is excluded; the old single-seed bound of ~0.008 shrinks fourfold. Criterion 4 ("keep the advantage with networks left out") is met at this prior.

- **Task 3 (part 1)** (Brief 9 point 2 on the old leave-out run: held vs seen with run as a random effect): none. On the old single-seed run (2 Nov, 500k, 4 groups) the held-minus-seen gap stays uninformative under every interval: +0.0028, with the REML interval [−0.0022, +0.0077]. Superseded by the three-seed 1M runs of Task 3 (12 runs, finished 8 Nov 18:19): +0.0004, REML [−0.0015, +0.0023].
- **Task 8, step 1** (TabPFN-3.5 licence (no training)): whether a TabPFN-3.5 fine-tuned on R1's prior is admissible as a rival. The weights' licence allows non-commercial research fine-tuning, but §2(d) forbids using TabPFN-3.5 outputs to train, fine-tune or distill a model competitive with it, and §3(d) forbids distributing or hosting derivatives. Fine-tuning TabPFN itself for an internal comparator looks allowed; publishing the fine-tuned weights does not.

Labels: **Verified** (file and row), **Reported** (stated by a source, not checked here), **Conjecture**. One page per task follows.

Task 1 — Item 0a: allocation types, near-copies, latent R^2, distinct contexts

Decisions for César: (1) the "allocation off" axis does not run: 0.06 % agnostic processes survive the verifier (rule: > 1 %); (2) the verifier rejects agnostic draws at zero confounding strength, so 3.5.2 should say the trained prior has practically no agnostic (U-driven) allocation; (3) DISTINCT=1 cannot make contexts distinct above the slice size (~780 rows): over 1,024 rows 99–100 % of contexts are longer than their slice.

Ran: commit 6cb459c (branch audit-brief9, 8fa4b74 + Brief 8 and 9 patches, clean tree), siena (RTX 5090, CPU only), --draws 5000 --batches 400 --batch-size 4, 7 Nov 21:36, pool md5 341e229d88f4 (E13a KCH, 1,207 rows, 1,207 distinct latent rows). Runs: final (context with replacement) and dist (DISTINCT=1).

Table (final):

- Allocation types, sampled processes: axis 0.609, severity 0.290, agnostic 0.101 (hyper weights 0.6 : 0.3 : 0.1, BIAS U(0, 1)).
- Before the verifier (1,814 attempts): axis 0.596, severity 0.285, agnostic 0.119. After (1,600 returned): axis 0.676, severity 0.323, agnostic 0.0006 (1 process). Pass rates: axis 1.000, severity 1.000, agnostic 0.0047. Returned after failing every attempt: 0.
- Agnostic pass rate by strength bin (BIAS tenths): 0/24, 0/18, 0/19, 0/19, 0/26, 1/23, 0/21, 0/14, 0/24, 0/27. Verifier statistic, median [p10, p90], tolerance 1e-3: 0.090 [0.033, 0.148] at 0.0–0.1; 0.017 [0.007, 0.032] at 0.4–0.5; 0.015 [0.002, 0.027] at 0.5–0.6 (zero strength); 0.096 [0.047, 0.129] at 0.9–1.0.
- Near-copies (10,046 virtual networks): cross-split halves of the same network 4.8 %; whole network against its own half 5.0 %.
- R^2 (5-fold ridge) on the latents: centroid x 0.934, y 0.928, z 0.940; log volume 0.947.
- Distinct context rows (final): slice median 773–790; distinct share 0.785 / 0.614 / 0.422 / 0.276 for context bins 256–511 / 512–1023 / 1024–2047 / 2048–3000, against 0.782 / 0.616 / 0.424 / 0.276 expected for draws with replacement. dist run: 1.000 / 0.991 / 0.493 / 0.286; contexts longer than their slice 13 % / 51 % / 99 % / 100 %. Agnostic pass rate under dist 0.010 (after: 0.13 %).

Reading:

- **Verified** (allocation_atoms_check.csv): axis and severity draws always pass; agnostic draws almost never pass, and the remaining share after the verifier is 0.06 %. Allocation towards the

better arm (agnostic) is, in practice, absent from training. Brief 9 point 4: no longer a conjecture on this pool and prior.

- **Verified** (allocation_by_type_and_strength.csv): at zero confounding strength (BIAS 0.4–0.6) the verifier statistic is still 15–17× its tolerance (median 0.015–0.017 vs 1e-3), and grows with strength. The rejection is a calibration effect of the in-sample logistic test on ~50 latents, not a targeted test (Brief 9 point 3). Axis and severity are tested for positivity only, so they are not "often rejected"; the trained prior is not closer to randomised allocation than 3.5.2 says on those two types.
- **Verified**: centroid and log volume are almost linear in the latents (R^2 0.93–0.95), so axis and severity allocation are confounding the model can see in X.
- **Verified** (distinct_context_rows_by_length.csv): on this pool the 3,000-row contexts have 27.6 % distinct rows, as draws with replacement from a ~780-row slice predict (Brief 9 point 6: Claude's ≈26 % is right for this pool). The 52 % of 2 Nov came from another pool (conjecture: the C1 prior's larger pool). Distinct contexts cap at the slice size (~780 rows), well below the 2,762-patient scored slice.
- **Conjecture**: since DISTINCT=1 changes only contexts shorter than ~1,000 rows, the 'distinct contexts' sweep level would mostly change the context-length distribution; report its results by context-size bin, as Brief 9 asks.

Deviations from this runbook: none. VENV=$HOME/env (the scaling runs' env).

Task 9 — Proximity check (stage 2 preparation)

Decisions for César: the threshold (fixed in advance by the plan). Two benchmark trials have a virtual network of the prior at $|r| \geq 0.9$; sixteen at ≥ 0.8.

Ran: commit 6cb459c (branch audit-brief9), siena CPU, --draws 10000 (10,000 processes, 19,983 virtual networks), 7 Nov 21:36, pool md5 341e229d88f4. The script's self-check of its vectorised τ against the DGP's cate() on the first 20 draws passed (no stop).

Table (proximity_summary.csv): median over the 32 trials of the max |corr| of τ with a virtual network 0.802; trials with a virtual network at $|r| \geq 0.9$: 2; at ≥ 0.8: 16.

Closest trials (proximity_by_trial.csv): genetics network 10 (948 enrolled) 0.951, 3 virtual networks ≥ 0.9, best agreement on enrolled 0.986; genetics network 13 (129 enrolled) 0.935, 1 network ≥ 0.9, agreement 0.977. Next: genetics 2 0.883, receptor 6 0.881, receptor 16 0.857.

Reading:

- **Verified** (files above): at 10,000 draws the prior produces effect vectors close to most benchmark trials (half at ≥ 0.8), and near-copies (≥ 0.9) for two genetics trials.
- **Conjecture**: these two trials are the ones where the in-prior/out-of-prior distinction matters most for stage 2; the decision depends on the threshold, not applied here.

Deviations from this runbook: none.

Task 5a — KCH split checks (Brief 9 point 8)

Decisions for César: whether a 604-lesion test half is acceptable. No B-half trial has fewer than 50 enrolled (minimum 50, median 142), but 14 of 32 B-half trials have a class (only-half-1 or only-half-2) below 10, against 8 of 32 in the whole pool; 6 of those classes are structurally 0 in the whole pool too.

Ran: commit 6cb459c (branch audit-brief9), siena CPU, split seed 0, 7 Nov, pool md5 341e229d88f4 (E13a KCH; cache channel disconnectome, threshold 0.5). Patient check with a file-name regex (kept on siena, not in the report).

Table (kch_split_check.csv, kch_trial_sizes.csv):

- 1,207 lesions; 1,207 distinct latent rows, 1,207 distinct overlap rows, 1,207 distinct patients (every file matched the regex): one lesion per patient, no exact repeats.
- Half A (train) 603 lesions, md5 b4104431c261; half B (test) 604 lesions, md5 57bb79b68be8. Both stay on siena.
- B half: enrolled min 50 / median 142; trials < 50 enrolled: 0; smallest: genetics 12 (50), receptor 5 (51), genetics 5 (52), receptor 12 (53), genetics 13 (65), receptor 13 (68).
- B-half trials with a class < 10 (only-half-1 / only-half-2): receptor 2 (0/43), 4 (0/80), 5 (6/18), 6 (93/0), 7 (0/95), 8 (0/15), 9 (81/6), 10 (6/163), 13 (35/2), 14 (7/112); genetics 5 (9/15), 8 (5/13), 9 (116/1), 13 (52/1).

Reading:

- **Verified** (kch_split_check.csv): the pool has one lesion per patient, so a split by lesion is a split by patient.
- **Verified** (kch_trial_sizes.csv): enrolment per trial is adequate in the B half (≥ 50), but the arms defined by which half is hit become thin: 14 trials have a class under 10, and several are already empty in the whole pool (a network half that is never hit alone).
- **Conjecture**: the empty classes are anatomical (one half of the network is nested in the other's territory) and would affect any KCH-based test, not only the split; prescriptive balanced accuracy will be undefined or noisy on those trials.

Deviations from this runbook: none (the --patient-regex was given; neither it nor any name appears in outputs).

Task 5b — R0 trained on KCH half A, scored on validation (Brief 9 point 8, task 5)

Decisions for César: whether to adopt the KCH split as the fresh test. Training R0 on half the pool (603 lesions) costs 0.0025 PEHE against final1M (interval [−0.0051, +0.0001], p 0.058). That is below the 0.005 screening rule and not significant, but the sign is consistent across splits. Together with 5a (thin hit-classes in 14 of 32 B-half trials), the split looks usable for PEHE but weak for prescriptive balanced accuracy.

Ran: commit 6cb459c (branch audit-brief9), sweep_axis.sh kchA, level R0, seeds 0, 1, 2, siena RTX 5090, 8 Nov 18:19 – 23:32 GMT; pool half A md5 b4104431c261, atlas md5 fd1b531e7cf9; checkpoints 225ca75355d9 / cfc5ea1e21c7 / 32cb5b2bd073, 0 error lines. Scored on validation (clean context, 0 of 1291 queries with a same-patient image in the training side), E13a seed 0 latents, score_val_clean.sh on CPU; stage1_screen.py. Nothing scored on half B.

Table (r0_half/kchA_vs_final1M.csv, md5 f63c7b57d61d), pehe_paper, 16 deficits, gain = final1M − kchA:

	receptor	genetics	mean of splits
kchA s0 / s1 / s2	0.296 / 0.295 / 0.300	0.260 / 0.268 / 0.267	

Mean, final1M → kchA			0.2788 → 0.2813
Gain	−0.0040	−0.0009	−0.0025 [−0.0051, +0.0001], p 0.058 (Holm 0.058)
Deficits where kchA is better			5 of 16
Screen	same sign on both splits: yes; passes 0.005 rule: no; claim bar 0.02: no		

Reading:

- **Verified** (kchA_vs_final1M.csv): halving the training pool costs at most ~0.005 PEHE on validation (upper end of the interval); the point estimate is 0.0025, about 1% of the PEHE level.
- **Verified** (score logs): the three seeds agree within 0.005 on each split; the cost is concentrated on receptor (−0.0040) rather than genetics (−0.0009).
- **Reported** (5a): half B has ≥ 50 enrolled in every trial, but 14 of 32 trials have a hit-class under 10.
- **Conjecture**: R0 depends weakly on pool size at this scale, so a model trained on half A is a fair stand-in for final1M when testing on half B; the test half will be informative for PEHE, noisy for balanced accuracy on the thin trials.

Deviations from this runbook: none (scoring on CPU with nice -n 10; the GPU was free before the launch).

Task 4 — Learning curve of the latent reference (validation only)

Decisions for César: none yet. On the only cells large enough to compare (20 matched cells: network 10 in both splits × 10 pseudo-folds), the reference is still falling between 2,000 patients and all, but by less each doubling; the full table is in learning_curve_summary_E1.csv.

Ran: commit 6cb459c (branch audit-brief9), siena, TabPFN 9.2.0 (model v3.5, ~/env_panel) on the GPU shared with the task 3 trainings, E1 seed 0 latents, validation pseudo-folds with clean context (0 of 1,291 queries with another image of the patient in the training side), sizes 500/1,000/2,000/all, 3 repeats, 7 Nov 21:47–22:52. Coverage cache recomputed from the disconnectome maps (kept on siena).

Table (learning_curve_summary_E1.csv), PEHE mean of splits (receptor / genetics):

- Matched cells with > 2,000 enrolled training patients (20 cells, median 2,604): n = 500 0.279 (0.278 / 0.280); 1,000 0.247 (0.237 / 0.257); 2,000 0.224 (0.213 / 0.235); all 0.216 (0.203 / 0.230).
- Cells with at least n (more cells at small n): 500 0.258 (230 cells); 1,000 0.234 (120); 2,000 0.224 (20); all 0.238 (320 cells, median 794 enrolled).

Reading:

- **Verified** (summary CSV): on the matched cells the latent reference keeps improving with more patients: −0.032 from 500 to 1,000, −0.022 from 1,000 to 2,000, −0.008 from 2,000 to all (~2,600).
- **Conjecture**: the reference is still somewhat sample-limited at the benchmark's size, with diminishing returns; the 0.242 (test) floor is not a hard floor. Only 20 matched cells, all from network 10 (receptor and genetics), support this, so the slope beyond 2,000 is uncertain.

- **Verified**: validation values are not comparable with the test 0.242/0.243 in level (runbook note); compare shapes only.

Deviations from this runbook: none in the commands. One CUDA allocation warning (OOM, then retry) at 21:57 while TabPFN shared the GPU with three trainings; all 1,430 cell fits completed without missing values and the trainings logged no error.

Task 2 — Wave 0 item 1: nine cells on validation, with pehe_implied

Decisions for César: (1) in the noisy cells, against the implied CATE TE(1 − RE)·τ, the final model is clearly worse than TabPFN-3.5 and CausalPFN (+0.10 to +0.17 PEHE): it does not shrink its effects as much as the noise requires, so the direction left open in Brief 8 (explanation 5) is "the final model overshoots"; (2) under location bias (0.95 and the default) the final model beats every estimator by 0.05–0.07, the result that supports the biased-cell criterion; (3) the primary biased cell to fix in the protocol (location bias 0.5 proposed) is the weakest one: at 0.5 TabPFN is still 0.014 better.

Ran: commit 6cb459c (branch audit-brief9); scorer ~/ncp_g8_b9 (99a3d8c + giles_replica from the Brief 8 patch, commit bdfd84b); per-fold code ~/ncp_pf_clean 79e8b3c; siena, 7–8 Nov 21:47–03:54; final1M seeds 0, 1, 2 (~/p2_bcb/pfn_final1M), E13a seed 0 latents; validation pseudo-folds with clean context. Gate: the patched scorer reproduces final1M seed 0 in the ideal cell exactly (difference 0.000 on both splits).

Table (wave0_tables.csv, wave0_tables_paired.csv): PEHE, mean of splits; d = final − other over 16 deficits; * = 95 % interval excludes 0. Metric: pehe_implied in the noisy cells, pehe_paper elsewhere.

Cell	Final (3 seeds)	TabPFN S	CausalPFN	LR (T)	ET (T)
noisy_te075	0.309	0.211 (+0.097*)	0.222 (+0.086*)	0.326 (−0.017)	0.251 (+0.058*)
noisy_te050	0.255	0.131 (+0.124*)	0.132 (+0.122*)	0.278 (−0.024*)	0.188 (+0.067*)
noisy_te025	0.213	0.042 (+0.171*)	0.050 (+0.162*)	0.248 (−0.035*)	0.148 (+0.064*)
location_bias_050	0.287	0.273 (+0.014*)	0.289 (−0.001)	0.316 (−0.029*)	0.363 (−0.076*)
location_bias_095	0.345	0.406 (−0.061*)	0.399 (−0.054*)	0.396 (−0.052*)	0.449 (−0.105*)
location_bias	0.348	0.421 (−0.072*)	0.401 (−0.053*)	0.398 (−0.050*)	0.454 (−0.106*)
responsiveness_060	0.282	0.274 (+0.008)	0.283 (−0.001)	0.313 (−0.030*)	0.361 (−0.079*)
responsiveness_075	0.300	0.294 (+0.006)	0.306 (−0.006)	0.326	0.383 (−0.083*)

				(−0.026*)		
responsiveness_090	0.358		0.367 (−0.008)	0.386 (−0.028*)	0.387 (−0.029*)	0.453 (−0.095*)

Against ±1 (pehe_paper), the noisy cells give final / TabPFN 0.500 / 0.470 (te075), 0.623 / 0.633 (te050), 0.692 / 0.688 (te025).

Reading:

- **Verified** (paired CSV): scored against the implied CATE, the final model's PEHE in the noisy cells is 0.21–0.31, against 0.04–0.22 for TabPFN and CausalPFN; the gap grows as TE falls (te025 largest). Against ±1 the estimators are close, as Table 4.7 reported.
- **Conjecture**: the final prior's virtual-trial effects are ±TE(1 − RE) with TE 0.9–1 and RE 0–0.05, so the model has never seen small effects and cannot shrink; the S-learners shrink because the outcomes are noisy. A wider TE range in the prior (the "wide ranges" factor B of stage 1) is the natural fix.
- **Verified**: under location bias at 0.95 and at the default strength the final model is better than every estimator (0.05–0.07 against TabPFN and CausalPFN); at 0.5 TabPFN is better by 0.014.
- **Verified**: in the responsiveness cells the final model is level with TabPFN and CausalPFN at 0.60 and 0.75, and better than CausalPFN at 0.90 (−0.028); it beats LR and ET everywhere.

Deviations from this runbook: wave0_cells.sh was not run as is, because on siena CausalPFN lives in ~/env_cpfn and TabPFN in ~/env_panel; the same calls (same labels and arguments) ran from ~/p2_bcb/brief9_t2_cells.sh with the matching env, two scorings at a time. TabPFN shared the GPU with the task 3 trainings (peak ~31 of 32.6 GB); no cell failed.

Task 6 — Stage 1 sweep: 2^(4−1) factorial and generic share 0.25 (24 runs), screened against R0

Decisions for César: (1) no level passes the screen, so under the preliminary protocol the next prior stays at R0 (final1M); (2) wide virtual-trial ranges (factor B) cost 0.013 PEHE, a robust harm in every run that has them; (3) generic share 0.75, effect term 0 and cosine decay are flat (|effect| ≤ 0.0014, none significant); (4) the preliminary protocol (docs/protocol_stage1.md, "preliminary") needs your review, the rotation-averaging point first.

Ran: commit cc83a62 (branch audit-brief9 = 6cb459c + preliminary protocol, md5 d9dd61f71a08, local, not pushed), stage1_jobs.sh, jobs F2–F8 and gw025 × seeds 0, 1, 2 (24 runs; `dist` dropped by the protocol), siena RTX 5090, 9 Nov 15:21 – 11 Nov 04:49 GMT, pool md5 341e229d88f4, atlas md5 fd1b531e7cf9; 24 checkpoints, 0 error lines. Scored on validation (clean context), E13a seed 0 latents, CPU; stage1_screen.py.

Table (stage1_screen.csv), pehe_paper, 16 deficits, gain = R0 − level (positive = better than R0), R0 mean 0.2788:

Level	Change	Gain receptor	Gain genetics	Gain mean [interval]	Holm p	Passes
F2	A + D	+0.0012	−0.0017	−0.0003 [−0.0024, +0.0019]	1.00	no
F3	B + D	−0.0129	−0.0150	−0.0140 [−0.0185,	< 0.001	no (worse)

				−0.0095]		
F4	A + B	−0.0155	−0.0124	−0.0139 [−0.0171, −0.0108]	< 0.001	no (worse)
F5	C + D	+0.0016	−0.0002	+0.0007 [−0.0017, +0.0032]	1.00	no
F6	A + C	−0.0041	−0.0039	−0.0040 [−0.0070, −0.0010]	0.048	no (worse)
F7	B + C	−0.0145	−0.0121	−0.0133 [−0.0173, −0.0093]	< 0.001	no (worse)
F8	A + B + C + D	−0.0155	−0.0123	−0.0139 [−0.0168, −0.0110]	< 0.001	no (worse)
gw025	generic 0.25	−0.0003	+0.0004	+0.0001 [−0.0026, +0.0027]	1.00	no

Main effects (stage1_screen_factorial.csv; 12 runs against 12, Holm over four), gain of the high level: A generic 0.75 −0.0014 [−0.0030, +0.0003]; B wide ranges −0.0129 [−0.0153, −0.0105], Holm p < 0.001; C CATE_W 0 −0.0006 [−0.0022, +0.0010]; D cosine +0.0010 [−0.0013, +0.0032]. Aliased pairs all |·| ≤ 0.0013, none significant.

Reading:

- **Verified** (stage1_screen.csv): no level reaches a 0.005 gain; the best point estimate is F5 at +0.0007. The protocol's combination rule therefore keeps R0.
- **Verified** (stage1_screen_factorial.csv): wide virtual-trial ranges are the only factor with an effect, and it is a harm of about 0.013 PEHE (about 5 % of the PEHE level), consistent across both splits and all four runs that contain B.
- **Verified**: generic share is flat between 0.25 (gw025), 0.5 (R0) and 0.75 (main effect A), within ±0.0015.
- **Conjecture**: the clean ranges match the benchmark's trial design, so widening them spends capacity on trials the benchmark never shows; the PFN at this prior and budget sits on a plateau for the other three factors.
- **Conjecture**: F6 (A + C, −0.0040, Holm p 0.048) is a marginal harm that the main effects do not explain (A and C are each flat); it may be the aliased AC = BD interaction (−0.0013, p 0.08) or noise.

Deviations from this runbook: `dist` not run (protocol section 5, after task 1); the protocol is preliminary and written before the sweep, not by César.

Task 3 — Leave-out at 1M with three seeds: held vs seen, run as a random effect (Brief 9 point 2)

Decisions for César: none needed to proceed. Leaving a network out of the prior costs nothing measurable on it: held − seen = +0.0004 PEHE, REML interval [−0.0015, +0.0023]. A penalty above ~0.002 for unseen networks is excluded; the old single-seed bound of ~0.008 shrinks fourfold. Criterion 4 ("keep the advantage with networks left out") is met at this prior.

Ran: commit 6cb459c (branch audit-brief9), lno3_launch.sh, 4 groups (networks 1–4, 5–8, 9–12, 13–16 left out of every channel; 60 atoms remain in each run) × seeds 0, 1, 2, 250,000 steps (1M trials), siena RTX 5090, 7 Nov 21:36 – 8 Nov 18:19, pool md5 341e229d88f4; 12 checkpoints, 0 error lines. Scored on validation (clean context), E13a seed 0 latents, lno3_score.sh; lno_cluster.py for the run effect.

Table (lno3_cluster.csv, paired/*.csv), pehe_paper, 12 runs × 16 deficits:

Contrast	Estimate	Naive paired t	Cluster mean (n = 12)	REML (run random effect)
Held − seen, both splits	+0.0004	[−0.0013, +0.0021], p 0.63	[−0.0019, +0.0027]	[−0.0015, +0.0023], se 0.0009
Held − seen, receptor	+0.0014	[−0.0015, +0.0044]	[−0.0025, +0.0054]	[−0.0017, +0.0045]
Held − seen, genetics	−0.0007	[−0.0033, +0.0020]	[−0.0026, +0.0013]	[−0.0034, +0.0021]

Mean PEHE held 0.2795, seen 0.2791. Against final1M (3 seeds, no network left out): held +0.0007 (t p 0.45), seen +0.0003 (t p 0.76). REML components (both): sd_run 0.0014, sd_resid 0.0060. Prescriptive balanced accuracy, both: held − seen +0.0023, REML [−0.0109, +0.0154] (2 deficits undefined).

Reading:

- **Verified** (lno3_cluster.csv): with three seeds at the final budget, a network the prior never contained is scored as well as one it contained: the held-minus-seen gap is +0.0004, and every interval excludes a penalty above 0.0027.
- **Verified** (paired CSVs): removing four networks from the prior does not cost the model anything against final1M either (held +0.0007, seen +0.0003, both n.s.).
- **Verified**: the run effect is small (sd 0.0014 against 0.0060 residual); the REML interval is close to the naive one, so Claude's clustering concern (point 2) changes little here.
- **Conjecture**: the virtual-trial prior teaches a mechanism (threshold rule on network halves) that transfers to unseen networks, rather than memorising the benchmark networks; this does not test mechanisms outside the family (the E4 effect-shape cell is still needed for that).
- **Verified**: the bound (0.0023) is about 3 % of the anatomy gain (~0.08, Table 4.9); as in task 3a, that gain comes from another configuration, so the comparison is indicative only.

Deviations from this runbook: none. The 12 trainings ran 3 at a time; the first batch shared the CPU with task 2 (≈7 h instead of ≈5 h).

Task 3 (part 1) — Brief 9 point 2 on the old leave-out run: held vs seen with run as a random effect

Decisions for César: none. On the old single-seed run (2 Nov, 500k, 4 groups) the held-minus-seen gap stays uninformative under every interval: +0.0028, with the REML interval [−0.0022, +0.0077]. Superseded by the three-seed 1M runs of Task 3 (12 runs, finished 8 Nov 18:19): +0.0004, REML [−0.0015, +0.0023].

Ran: commit 6cb459c (branch audit-brief9), lno_cluster.py on the 2 Nov scorings (labels lno_g{1..4}_s0, validation, clean context), siena CPU, 7 Nov. Pipeline check: the naive interval reproduces the reported

+0.0028 [−0.0012, +0.0067] exactly.

Table (lno_old_cluster.csv), pehe_paper, held − seen, 4 runs × 16 deficits:

Split	Estimate	Naive paired t	Cluster mean (n = 4)	REML (run random effect)
Both	+0.0028	[−0.0012, +0.0067], p 0.15	[−0.0058, +0.0113], p 0.38	[−0.0022, +0.0077], se 0.0016
Receptor	+0.0021	[−0.0016, +0.0058]	[−0.0058, +0.0100]	[−0.0027, +0.0069]
Genetics	+0.0034	[−0.0023, +0.0092]	[−0.0081, +0.0149]	[−0.0043, +0.0111]

REML components (both): sd_run 0.0033, sd_resid 0.0063. Prescriptive balanced accuracy, both: held − seen −0.0119, REML [−0.0339, +0.0140] (2 deficits undefined).

Reading:

- **Verified** (lno_old_cluster.csv): accounting for the shared run widens the interval from [−0.0012, +0.0067] to [−0.0022, +0.0077]; the estimate does not change. Claude's concern (point 2) holds in direction for the t-based REML interval, but the change is small.
- **Verified**: the upper bound is about 0.008, a tenth of the anatomy gain (~0.08 in Table 4.9), so a large region-specific share of that gain stays excluded, as Claude argued; but the gain and the leave-out runs differ in configuration (anatomy family present, E1 latents vs virtual-trial family, E13a), so this is a bound on this prior only (conjecture for the transfer).
- **Verified**: the run effect is half the residual spread (0.0033 vs 0.0063): runs do shift all deficits, but most of the spread is per deficit.

Deviations from this runbook: none.

Task 8, step 1 — TabPFN-3.5 licence (no training)

Decisions for César: whether a TabPFN-3.5 fine-tuned on R1's prior is admissible as a rival. The weights' licence allows non-commercial research fine-tuning, but §2(d) forbids using TabPFN-3.5 outputs to train, fine-tune or distill a model competitive with it, and §3(d) forbids distributing or hosting derivatives. Fine-tuning TabPFN itself for an internal comparator looks allowed; publishing the fine-tuned weights does not.

Ran: siena, ~/env_panel (the venv of the dissertation's "TabPFN-3.5" scorings: logs read "tabpfn 9.2.0: model=v3.5"), 7 Nov. No training, no model download.

Facts:

- pip show tabpfn: version 9.2.0, License-Expression Apache-2.0, author Prior Labs. The package's dist-info LICENSE is the Apache License 2.0 (202 lines). **Verified** (dist-info).
- Checkpoint in use: tabpfn-v3.5-20260909.safetensors (876,027,932 bytes, ~/.cache/tabpfn), from Hugging Face repo Prior-Labs/tabpfn_3_5 (model_loading.py, get_v3_5). Download requires a one-time licence acceptance (browser_auth.py). **Verified** (files).
- The weights carry a separate licence, "tabpfn-3-5-license-v1.0" (model card on huggingface.co/Prior-Labs/tabpfn_3_5): permissive for research and internal evaluation; "the model, its derivatives, and its outputs cannot be used for any commercial or production purpose". **Reported** (model card, read through a web fetch).
- Clauses (LICENSE file of the HF repo, read through a web fetch tool that summarises; quote to be

checked against the full text before it is cited). **Reported**:

- §1(a) Derivative: "(including but not limited to any customized, fine-tuned, retrained, or otherwise adapted version thereof)".
- §1(c) Non-Commercial Purpose: "use for testing, evaluation, or research not tied to commercial gain, production deployment, or revenue generation"; "This includes internal benchmarking, academic research, and experimentation on private or public datasets"; not non-commercial: "(c) to train, fine-tune, or distill other models for commercial use".
- §2(a) Grant: "solely for your Non-Commercial Purposes".
- §2(d) Outputs: "You may not use Outputs to train, fine-tune, or distill a model that is competitive with the TABPFN-3.5 Model."
- §3(c): a modified model must carry "a statement that you have modified the applicable TABPFN-3.5 Model".
- §3(d) No Hosted Service: "You may not Distribute, host, or make available the TABPFN-3.5 Model or any Derivative".

Reading:

- **Verified**: the Apache-2.0 licence covers the tabpfn Python package only, not the v3.5 weights.
- **Reported**: research fine-tuning of the weights is within the grant (§1(a), §1(c), §2(a)); distribution of fine-tuned weights is not (§3(d)).
- **Conjecture**: §2(d) concerns training another model on TabPFN's outputs; a Neuro-CausalPFN trained on its own prior does not use TabPFN outputs, so it is not affected, but any distillation from TabPFN predictions would be. César (or Prior Labs) should confirm before step 2.

Deviations from this runbook: the HF licence text was read through a summarising fetch, not printed verbatim; read the full text once before quoting it in the paper.